Induction Game (IG)

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What does induction mean in this **context**?

- The process of **generating** an abstract idea / program based on **some concrete evidence** (I/O pairs).
- Is there proof that this method works?
- [Is Programming by Example Solved by LLMs?]
- [CodeARC: Benchmarking Reasoning Capabilities of LLM Agents for Inductive Program Synthesis]

DOMAIN: lists

provided examples

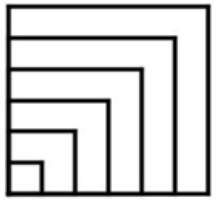
INPUT	OUTPUT
4,2,8	2,0,6
9,9,9,9	0,0,0,0
-7,0,2	0,7,9

generated program

```
# Check if the list is empty
if not input_list:
    return input_list
# Find min number in the list
min_num = min(input_list)
# Subtract from each element
return [num - min_num
        for num in input_list]
```

DOMAIN: graphics

provided example



generated program

```
for i in range(7):
    with fork_state():
        for j in range(4):
            forward(2*i)
            left(90.0)
```

DOMAIN: text editing macros

provided examples

INPUT	OUTPUT
18:25:57	6PM to 8PM
21:44:40	8PM to 10PM
07:00:20	6AM to 8AM
23:34:17	10PM to 12AM

generated program

```
original_time = datetime.strptime(input_str, '%H:%M:%S')
hour = original_time.hour
start_hour = hour - (hour % 2)
end_hour = start_hour + 2
start_hour_12 = start_hour % 12 or 12
end_hour_12 = end_hour % 12 or 12
start_ampm = "AM" if start_hour < 12 else "PM"
end_ampm = "AM" if end_hour < 12 or end_hour == 24 else "PM"
return f"{start_hour_12}{start_ampm} to {end_hour_12}{end_ampm}"
```

Figure 1: Domains, including standard ones that resemble programs found in pretraining data, as well as a less common graphics domain, which is likely less represented in LLM pretraining data.

Findings of the papers?

- Pretrained and Instruction tuned LLM’s are not good at induction / Programming by Examples.
- Fine tuning improves their performance significantly.

My interest in reading these papers

- LLM’s can do induction.
- Planning and making a roadmap.
- See some relevant material to programming with induction.

Induction Game Concept

Programmatic Player

Player 1

→ Give me two numbers.

1 , 1 ←

→ The output is 2

```
def add_two_numbers(a, b):
    return a + b
```

←

We are interested in the **output** of the function "*Guessed*" by player 1.

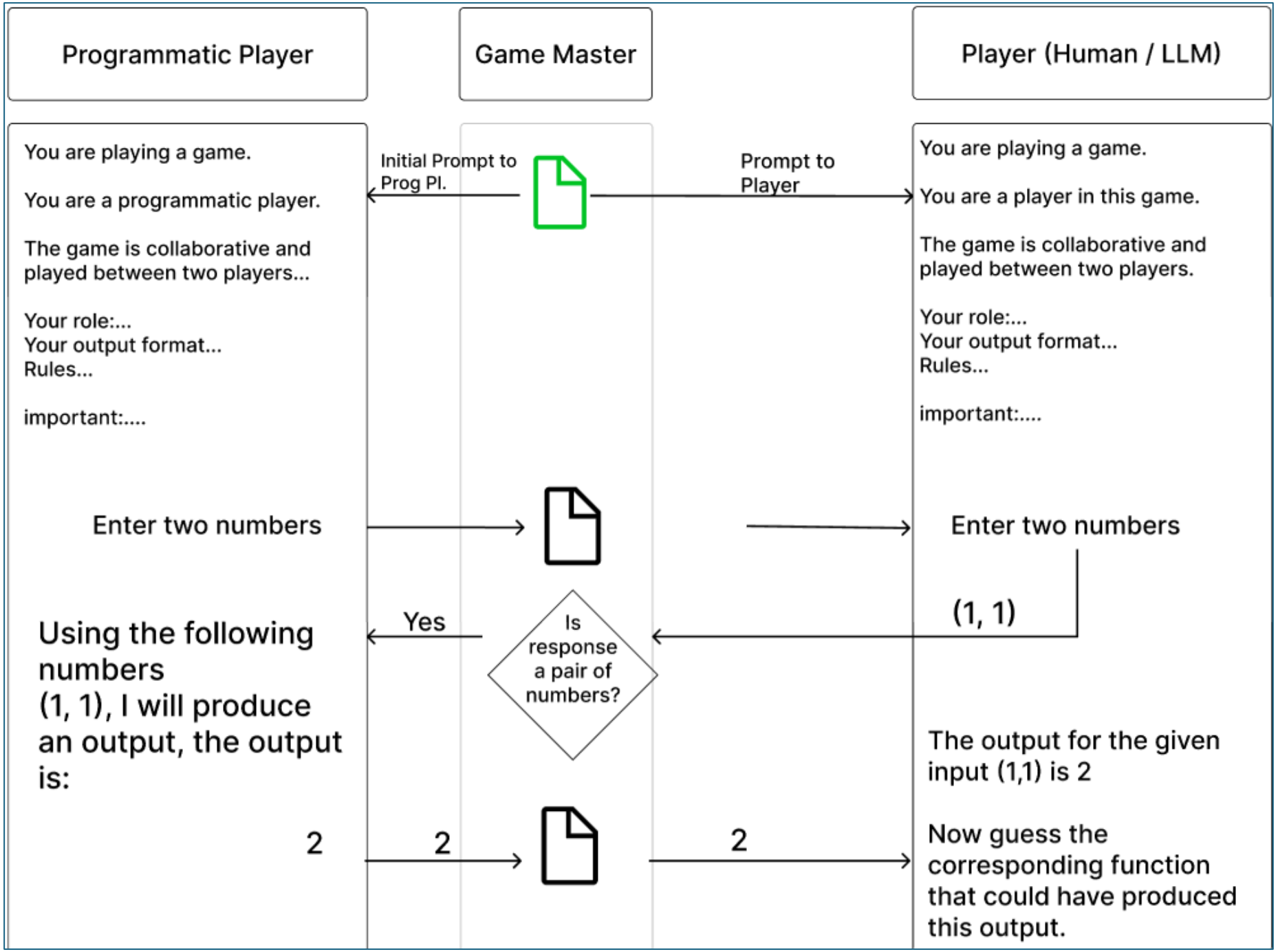
What **would** be require?

- I/O Samples
- The correct program
- Test cases to test the player 1’s program

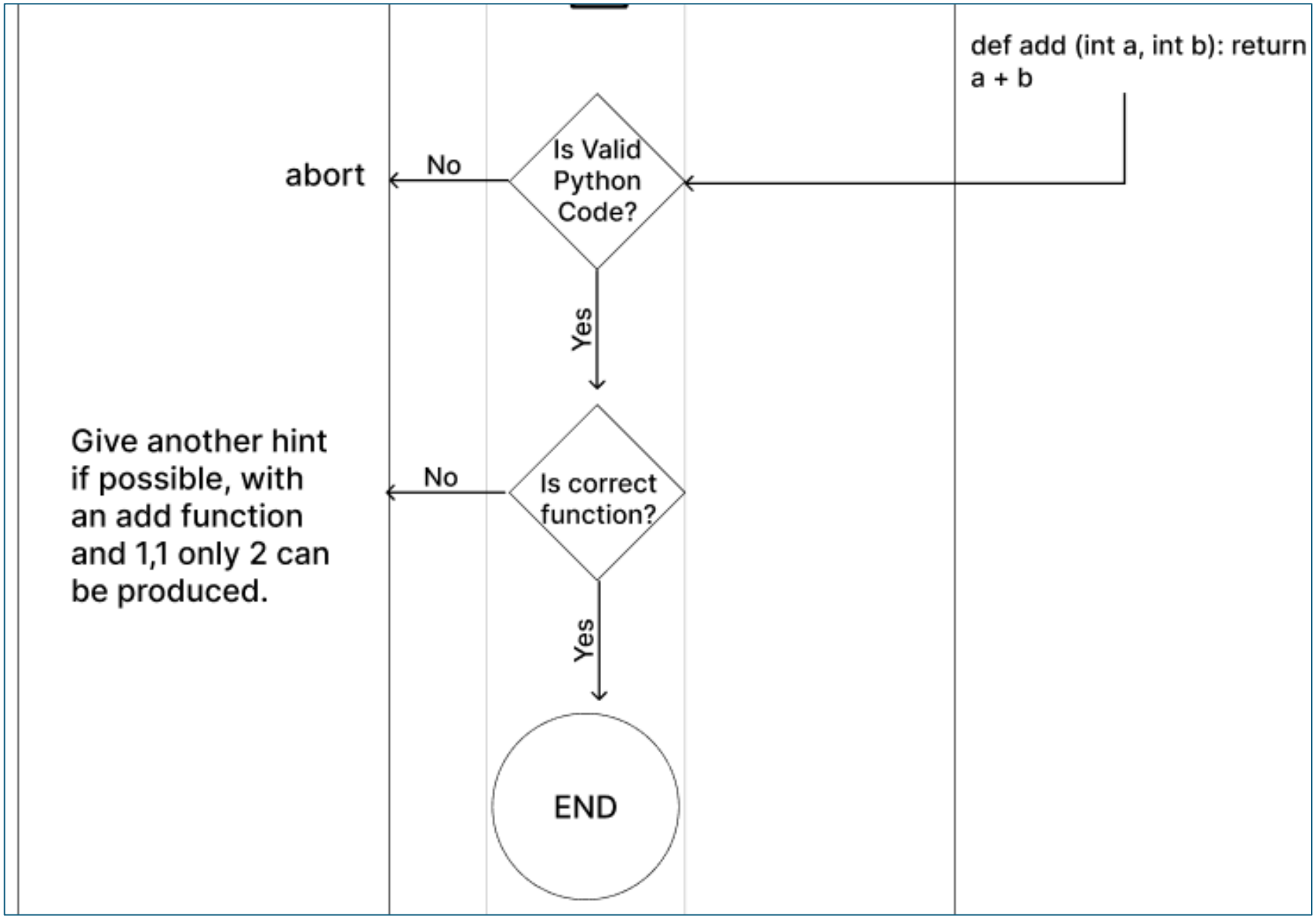
What I have seen so far...

Dataset Name	Focus	Key Features
CodeARC	General-Purpose Python Induction	dataset of **1114 diverse Python functions** with initial I/O examples.
DSCodeBench	Data Science Code Generation	has **Ground Truth Code** and an automatically generated **Test Case Script** for producing tailored input-output examples.
Program Synthesis Benchmark Suite (PSB)	Standard Algorithmic Problems	Provides canonical datasets (PSB1 and PSB2) for general programming tasks (e.g., fizz-buzz, gcd) with over **1 million labeled cases per problem**
Case2Code	Scalable Synthetic Data	Given a **function**, generates a pair of **inputs** and **outputs** for this function. Good if I want to generate large sets of I/O pairs.

Game Diagram



Game Diagram cont.



Timeline:

"If I had 8 hours to chop down a tree, I would spend 6 sharpening my knife"

Phase	Estimated Duration	Deliverables Date (DD.MM.YY)	Key Activities
Groundwork & Core Logic	1–2 Weeks	02.12.25	Reading papers, additional groundwork, and implementing **basic game logic** and player interactions.
Prompt Engineering	1 Week	09.12.25	Designing, testing, and refining the **LLM prompt templates** for the Induction Player.
Instances & Experiments	1 Week	16.12.25	**Finalizing game instances** (function set) and experiment parameters for initial runs.
Evaluation Metrics	1–2 Weeks	30.12.25	Identifying and implementing **evaluation metrics** (e.g., accuracy, queries needed).
Final Polishing	2–3 Weeks	20.01.25	Polishing and finalizing the **game setup. + Documentation.**

Related Work

- CodeARC: Benchmarking Reasoning Capabilities of LLM Agents for Inductive Program Synthesis
- Is Your Code Generated by ChatGPT Really Correct?
- Case2Code Dataset link
- Case2Code: Scalable Synthetic Data for Code Generation

Questions

- Are the functions kept as JSON?
- How many functions should there be?
- How can this morph into medical stuff?